

Lab 4: IP Addressing for Secure Transmission

Task 1: Identify IP Classes and Default Subnet Masks

The class of an IP address is decided by its first number (first octet). Class A is 1 to 126, Class B is 128 to 191, and Class C is 192 to 223.

- **10.45.3.29**: first octet is 10, so it is **Class A**. Default subnet mask: **255.0.0.0 (/8)**. (It is also a private address.)
- **172.20.5.10**: first octet is 172, so it is **Class B**. Default subnet mask: **255.255.0.0 (/16)**. (It is also a private address.)
- **192.168.1.25**: first octet is 192, so it is **Class C**. Default subnet mask: **255.255.255.0 (/24)**. (It is also a private address.)
- **122.121.120.119**: first octet is 122, so it is **Class A**. Default subnet mask: **255.0.0.0 (/8)**. (It is a public address.)

Task 2: Find Network and Broadcast Address

Method: Use the default mask of the class. The network address has all host bits as 0. The broadcast address has all host bits as 1.

1. 192.168.1.100

- Class C, mask 255.255.255.0 (/24). The first three numbers are the network part.
- **Network address:** 192.168.1.0
- **Broadcast address:** 192.168.1.255
- Usable range: 192.168.1.1 to 192.168.1.254

2. 11.12.13.14

- Class A, mask 255.0.0.0 (/8). Only the first number is the network part.
- **Network address:** 11.0.0.0
- **Broadcast address:** 11.255.255.255
- Usable range: 11.0.0.1 to 11.255.255.254

3. 10.0.5.50

- Class A, mask 255.0.0.0 (/8).
- **Network address:** 10.0.0.0
- **Broadcast address:** 10.255.255.255
- Usable range: 10.0.0.1 to 10.255.255.254

4. Roll number IP address (roll number 066)

The IP address made from the roll number is **66.66.66.66**.

- First octet is 66, so it is Class A, mask 255.0.0.0 (/8).
- **Network address:** 66.0.0.0
- **Broadcast address:** 66.255.255.255

- Usable range: 66.0.0.1 to 66.255.255.254

Quick rule for your own roll number: if it is Class A, keep the first number and make the rest 0 (network) or 255 (broadcast). For Class B, keep the first two numbers. For Class C, keep the first three numbers.

Task 3: Usable Networks and Hosts in Class B, C and D

Formulas: Number of networks = 2 to the power of network bits. Hosts per network = 2 to the power of host bits, minus 2. We subtract 2 because one address is the network address and one is the broadcast address, and these cannot be given to devices.

Class B

- Network bits: 16 total, but the first 2 bits are fixed (10), so 14 bits are free.
- **Number of networks:** $2^{14} = 16,384$
- Host bits: 16.
- **Usable hosts per network:** $2^{16} - 2 = 65,534$

Class C

- Network bits: 24 total, but the first 3 bits are fixed (110), so 21 bits are free.
- **Number of networks:** $2^{21} = 2,097,152$
- Host bits: 8.
- **Usable hosts per network:** $2^8 - 2 = 254$

Class D

Class D is used for **multicasting** (sending data to a group of devices). It is not divided into network and host parts, so it has **no usable networks and no usable hosts** for normal devices. The first bits are fixed as 1110, which leaves 28 bits, so there are $2^{28} = 268,435,456$ multicast group addresses (224.0.0.0 to 239.255.255.255).

Task 4: Private IP Address Ranges and Their Role in Security

Private IP ranges

- **Class A:** 10.0.0.0 to 10.255.255.255 (10.0.0.0/8)
- **Class B:** 172.16.0.0 to 172.31.255.255 (172.16.0.0/12)
- **Class C:** 192.168.0.0 to 192.168.255.255 (192.168.0.0/16)

Private addresses are used only inside a home, school, or company network. They cannot be used directly on the Internet.

Role in secure information systems

- **Hidden from the Internet:** Outside users cannot reach a private IP directly, so attackers cannot easily find or attack internal devices.
- **Works with NAT:** The router changes private addresses into one public address, so the inside network stays hidden.

- **Easy separation:** Different departments can use different private ranges. This keeps their data separate and lets us apply firewall rules and access lists between them.
- **Less attack area:** Fewer devices are exposed to the Internet, so there are fewer chances of attack.
- **Saves public IPs:** Thousands of devices can share one public address.
- **Safe testing:** Labs and test systems can be built without touching the real Internet.

Task 5: Secure Network Design for Gulberg Campus

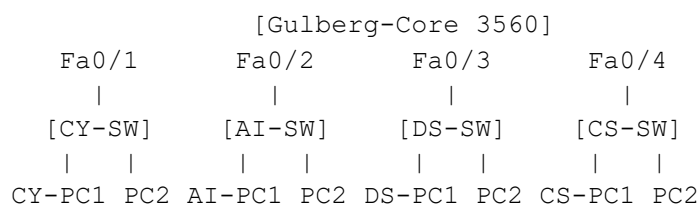
Design idea

There are four departments: CY, AI, DS, and CS. Each department has its own switch and its own network. All four switches connect to one central core switch (a multilayer switch) that routes data between the departments. This way, every department can share data with every other department. Every IP address starts with the roll number (066).

Devices used in Packet Tracer

- 1 x Core switch: Multilayer switch 3560 (named Gulberg-Core)
- 4 x Department switches: 2960 (CY-SW, AI-SW, DS-SW, CS-SW)
- 8 x PCs: two in each department
- Cables: Copper Straight-Through everywhere

Topology



IP addressing plan (roll number 066)

- **CY department:** Network 66.1.1.0/24. Gateway 66.1.1.1. PCs: 66.1.1.10 and 66.1.1.11
- **AI department:** Network 66.1.2.0/24. Gateway 66.1.2.1. PCs: 66.1.2.10 and 66.1.2.11
- **DS department:** Network 66.1.3.0/24. Gateway 66.1.3.1. PCs: 66.1.3.10 and 66.1.3.11
- **CS department:** Network 66.1.4.0/24. Gateway 66.1.4.1. PCs: 66.1.4.10 and 66.1.4.11
- Subnet mask for all PCs: 255.255.255.0

The gateway is the address of the core switch port that faces that department. Each PC uses it to send data to other departments.

Step 1: Build the network

- Place the core switch, four 2960 switches, and eight PCs.

- Connect each 2960 switch (port GigabitEthernet0/1) to the core switch: CY to Fa0/1, AI to Fa0/2, DS to Fa0/3, CS to Fa0/4.
- Connect each PC to its department switch on Fa0/1 and Fa0/2.
- Wait until all link lights are green.

Step 2: Configure the core switch (Gulberg-Core)

Click the core switch, open the **CLI** tab, and type the commands below one small group at a time. Press Enter after each line.

2.1 — Go into setup mode and give it a name

```
enable      configure      terminal
hostname    Gulberg-Core   enable
secret cisco123
```

This unlocks setup mode, renames the switch to Gulberg-Core, and sets an admin password (cisco123) so no one else can change the settings.

2.2 — Turn on routing (so departments can reach each other)

```
ip routing
```

Without this one line, the four departments would stay separate and could not share data. This is the most important command in the whole lab.

2.3 — Give the CY port its own gateway address

```
interface fastEthernet0/1
description Link to CY department
no switchport
ip address 66.1.1.1 255.255.255.0
no shutdown
exit
```

This opens port Fa0/1, turns it into a routed port ("no switchport"), and gives it the address 66.1.1.1. This address becomes the gateway for every CY-PC.

2.4 — Give the AI port its own gateway address

```
interface fastEthernet0/2
description Link to AI department
no switchport
ip address 66.1.2.1 255.255.255.0
no shutdown
exit
```

Same idea as before, but this is port Fa0/2 for the AI department, with gateway 66.1.2.1.

2.5 — Give the DS port its own gateway address

```
interface fastEthernet0/3
description Link to DS department
no switchport
ip address 66.1.3.1 255.255.255.0
no shutdown
exit
```

This is port Fa0/3 for the DS department, with gateway 66.1.3.1.

2.6 — Give the CS port its own gateway address

```
interface fastEthernet0/4
```

```

description Link to CS department
no switchport
ip address 66.1.4.1 255.255.255.0
no shutdown
exit

```

This is port Fa0/4 for the CS department, with gateway 66.1.4.1. All four departments now have a gateway on the core switch.

2.7 — Add a password and a warning message, then save

```

line console 0
password cisco123
login
exit
banner motd #Authorized Access Only#
end write memory

```

This asks for a password before anyone can use the console, shows a warning message to unauthorized users, and "write memory" saves the configuration so it is not lost.

Step 3: Configure each department switch

Now click the **CY-SW** switch and open its CLI tab. Type the same small groups as below. Afterwards, repeat all of Step 3 on AI-SW, DS-SW, and CS-SW — the only thing you change each time is the hostname (CY-SW becomes AI-SW, DS-SW, or CS-SW).

3.1 — Go into setup mode and give it a name

```

enable      configure      terminal
hostname CY-SW enable secret
cisco123

```

Same first step as the core switch: unlock setup mode, set the name, and set the admin password.

3.2 — Lock each PC port to one device

```

interface range fastEthernet0/1 - 2
switchport mode access switchport port-
security switchport port-security maximum
1 switchport port-security mac-address
sticky switchport port-security violation
restrict
exit

```

This picks both PC ports at once (Fa0/1 to Fa0/2) and turns on port security, so only 1 known device (learned automatically with "sticky") is allowed on each port. If a stranger plugs in another device, that port simply blocks it ("restrict") instead of shutting down.

3.3 — Add a warning message and save

```

banner
motd #Authorized Access Only# end
write memory

```

This shows the warning message and saves the settings, exactly like on the core switch.

Step 4: Set the IP address on every PC

Do this on every PC, one at a time:

- Click the PC.
- Open the **Desktop** tab.

- Click **IP Configuration**.
- Select **Static**.
- Type the IP Address, Subnet Mask, and Default Gateway for that PC (from the addressing plan above).

Example for CY-PC1:

```
IP Address       : 66.1.1.10
Subnet Mask      : 255.255.255.0
Default Gateway  : 66.1.1.1
```

Repeat for all 8 PCs. The gateway of a PC is always the .1 address of its own department (66.1.1.1 for CY, 66.1.2.1 for AI, 66.1.3.1 for DS, 66.1.4.1 for CS).

Step 5: Test the network (no data loss)

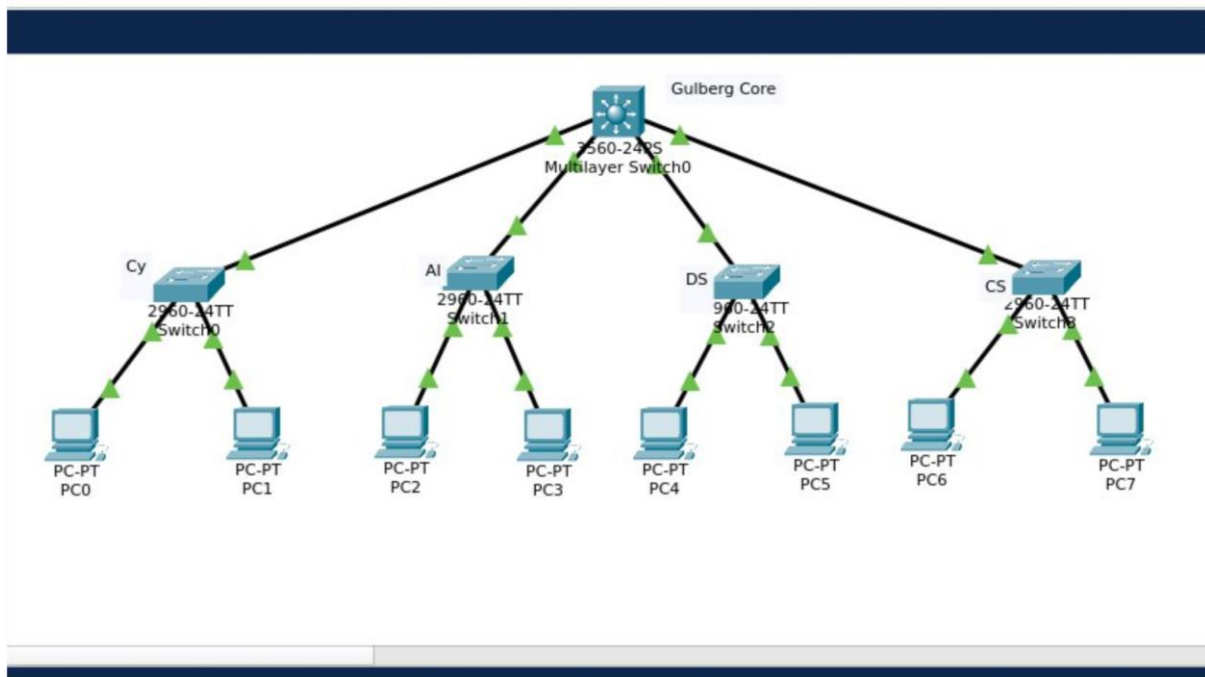
From CY-PC1, open Command Prompt and ping a PC in every other department:

```
ping 66.1.2.10
ping 66.1.3.10
ping 66.1.4.10
```

On the core switch, check the setup with these commands:

```
show ip interface brief
show ip route
```

Expected result: All pings show "Reply from..." with **0% loss**. This proves that data moves between all four departments without any loss. The route table shows four connected networks (66.1.1.0, 66.1.2.0, 66.1.3.0, 66.1.4.0). The first ping may time out once because of ARP. This is normal.

Screenshot 1: Complete topology**Screenshot 2: Core switch configuration (Gulberg-Core)**

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- FastEthernet0/1
- FastEthernet0/2
- FastEthernet0/3
- FastEthernet0/4
- FastEthernet0/5
- FastEthernet0/6
- FastEthernet0/7

Global Settings

Display Name: Multilayer Switch0

Hostname: Gulberg-Core

NVRAM: Erase Save

Startup Config: Load... Export...

Running Config: Export... Merge...

Device Clock: 01:14:30 Mon Mar 1 1993 UTC

Equivalent IOS Commands

```
Press RETURN to get started!
```

```
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
```

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Screenshot 3: Department switch configuration

Physical **Config** CLI Attributes

GLOBAL
 Settings
 Algorithm Settings
SWITCHING
 VLAN Database
INTERFACE
 FastEthernet0/1
 FastEthernet0/2
 FastEthernet0/3
 FastEthernet0/4
 FastEthernet0/5
 FastEthernet0/6
 FastEthernet0/7
 FastEthernet0/8
 FastEthernet0/9
 FastEthernet0/10

Global Settings
 Display Name: Switch0
 Hostname: CS-SW
 Serial Number: Serial Number
 NVRAM: Erase Save
 Startup Config: Load... Export...
 Running Config: Export... Merge...
 Device Clock: 01:21:39 Mon Mar 1 1993 UTC

Equivalent IOS Commands

```

CS-SW(config-if-range)#exit
CS-SW(config)#
CS-SW(config)#banner motd #Authorized Access Only#
CS-SW(config)#
CS-SW(config)#end
CS-SW#
CS-SW#write memory
Building configuration...
[OK]
CS-SW#
CS-SW#
%SYS-5-CONFIG_I: Configured from console by console
  
```

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Screenshot 4: IP configuration of CY-PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 66.1.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 66.1.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::201:96FF:FEB7:2C01

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

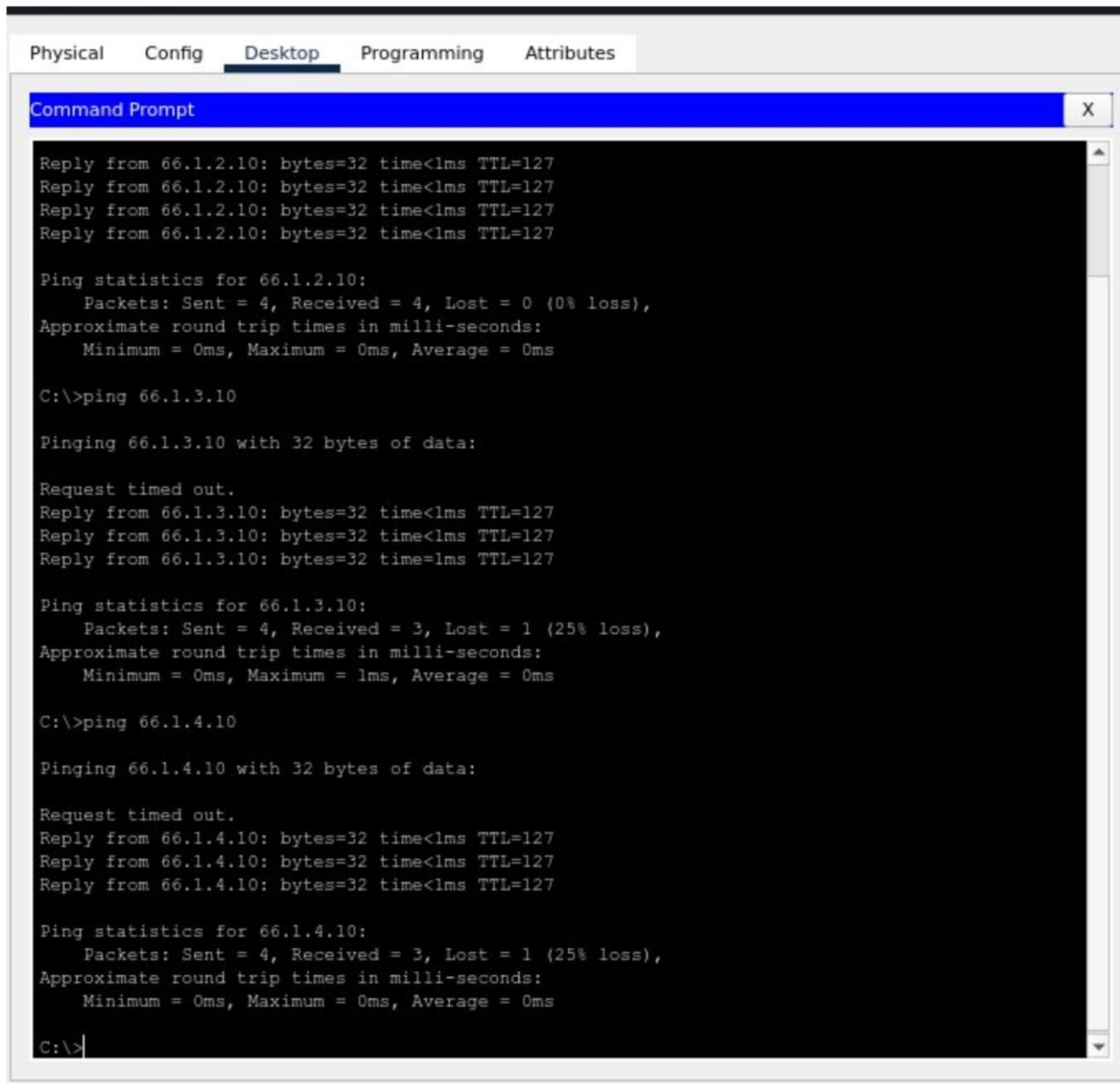
Authentication: MDS

Username:

Password:

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Screenshot 5: Ping from CY-PC1 to AI, DS, and CS



The screenshot shows a Cisco Packet Tracer interface with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the results of three ping commands. The first command is for 66.1.2.10, which shows successful replies with 0% loss. The second command is for 66.1.3.10, which shows a 25% loss (1 packet lost). The third command is for 66.1.4.10, which also shows a 25% loss (1 packet lost). The Command Prompt window has a title bar with 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes' tabs, and a close button 'X'.

```
Physical  Config  Desktop  Programming  Attributes

Command Prompt X

Reply from 66.1.2.10: bytes=32 time<1ms TTL=127
Reply from 66.1.2.10: bytes=32 time<1ms TTL=127
Reply from 66.1.2.10: bytes=32 time<1ms TTL=127
Reply from 66.1.2.10: bytes=32 time<1ms TTL=127

Ping statistics for 66.1.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 66.1.3.10

Pinging 66.1.3.10 with 32 bytes of data:

Request timed out.
Reply from 66.1.3.10: bytes=32 time<1ms TTL=127
Reply from 66.1.3.10: bytes=32 time<1ms TTL=127
Reply from 66.1.3.10: bytes=32 time=1ms TTL=127

Ping statistics for 66.1.3.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 66.1.4.10

Pinging 66.1.4.10 with 32 bytes of data:

Request timed out.
Reply from 66.1.4.10: bytes=32 time<1ms TTL=127
Reply from 66.1.4.10: bytes=32 time<1ms TTL=127
Reply from 66.1.4.10: bytes=32 time<1ms TTL=127

Ping statistics for 66.1.4.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Screenshot 6: show ip interface brief on the core switch

```

IOS Command Line Interface

%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Press RETURN to get started!

Authorized Access Only

User Access Verification

Password:

Gulberg-Core>show ip interface
FastEthernet0/1 is up, line protocol is up
  Internet protocol processing disabled
FastEthernet0/2 is up, line protocol is up
  Internet protocol processing disabled
FastEthernet0/3 is up, line protocol is up
  Internet protocol processing disabled
FastEthernet0/4 is up, line protocol is up
  Internet protocol processing disabled
FastEthernet0/5 is down, line protocol is down
  Internet protocol processing disabled
FastEthernet0/6 is down, line protocol is down
  Internet protocol processing disabled
FastEthernet0/7 is down, line protocol is down
  Internet protocol processing disabled
FastEthernet0/8 is down, line protocol is down
  Internet protocol processing disabled
FastEthernet0/9 is down, line protocol is down
  Internet protocol processing disabled
FastEthernet0/10 is down, line protocol is down
  Internet protocol processing disabled
FastEthernet0/11 is down, line protocol is down
  Internet protocol processing disabled
--More--

```

Screenshot 07: show ip route on the core switch

```

Gulberg-Core>
Gulberg-Core>
Gulberg-Core>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  66.0.0.0/24 is subnetted, 4 subnets
C      66.1.1.0 is directly connected, FastEthernet0/1
C      66.1.2.0 is directly connected, FastEthernet0/2
C      66.1.3.0 is directly connected, FastEthernet0/3
C      66.1.4.0 is directly connected, FastEthernet0/4

Gulberg-Core>

```

Conclusion

In this lab, IP classes, subnet masks, network and broadcast addresses, usable hosts, and private IP ranges were studied. A secure campus network was designed in which four departments use separate networks that start with the roll number and share data through a core switch without any loss. Passwords and port security were added to protect the network.